Supplementary Material: Full Proofs for Cost- and SLA-Bounded Orchestration for LLM-Agent Tool/Service Composition

Anonymous

Abstract

This supplement collects the formal problem statement and all theorem-like claims from the paper source under `paper/latex/sections`, with the corresponding full proofs or proof sketches preserved in LaTeX form. It is meant to compile to the root-level `supplement.pdf` referenced by the repository README.

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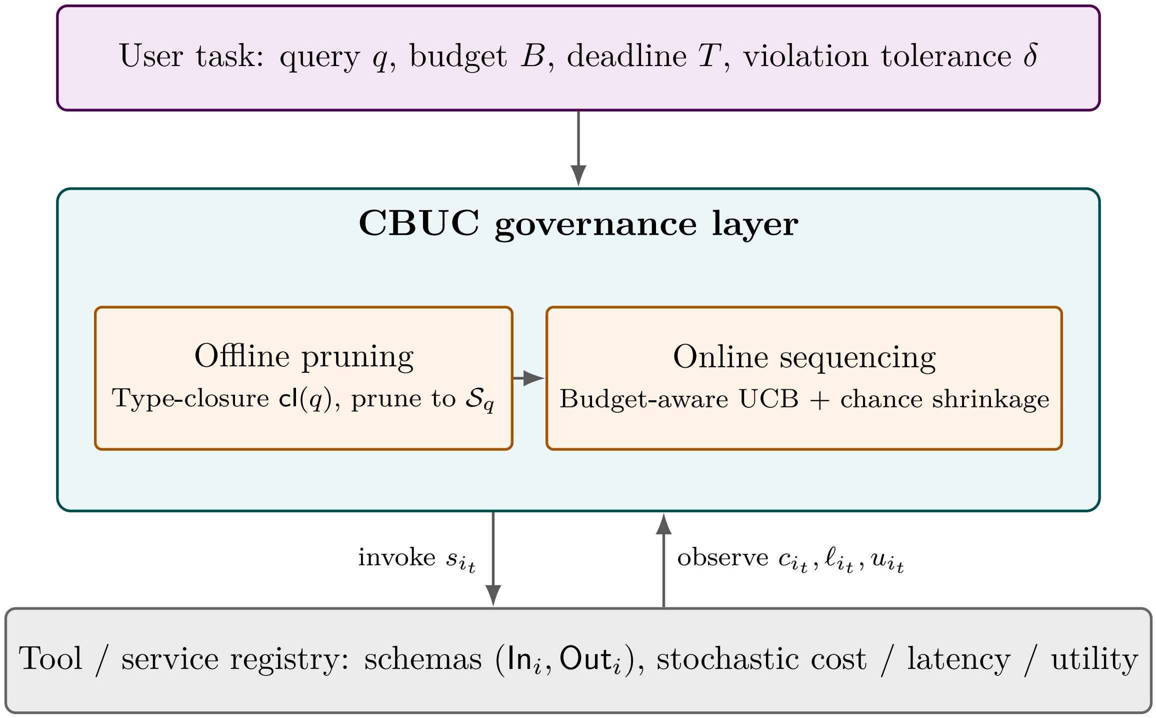

Figure 1: The CBUC governance layer mediates between the user task specification and the tool registry. Offline schema-type pruning and online chance-constrained UCB sequencing are its two phases; CBSLAO formalises everything that crosses the boundary between the layer and the registry.

1 Problem Formulation

1.1 Setting

Let $\mathcal{S} = \{s_1, \ldots, s_n\}$ be a *tool pool*. Each tool s_i exposes a schema $\sigma_i = (\mathsf{In}_i, \mathsf{Out}_i)$ over a common type universe $\mathcal{U}$. A *task* is a tuple $\tau = (\mathsf{goal}, q, B, T, \delta)$ where q is a query, $B \in \mathbb{R}_{>0}$ is a monetary/token budget, $T \in \mathbb{R}_{>0}$ is a wall-clock deadline, and $\delta \in (0,1)$ is an acceptable violation probability.

At each decision epoch $t = 1, 2, \ldots$, the agent holds a state $h_t \in \mathcal{H}$ (history of prior calls and observations) and selects either a tool s_i with a call payload or a STOP action emitting a final answer. An invocation of s_i draws

- cost $c_{i,t} \sim \mathcal{D}_i^c$;
- latency $\ell_{i,t} \sim \mathcal{D}_i^\ell$;
- an observation $o_{i,t}$ that updates the state: $h_{t+1} = h_t \oplus (s_i, o_{i,t})$.

A *utility function* $U : \mathcal{H} \to [0,1]$ measures the expected quality of the final answer given the history at STOP. The regret analysis in §3 restricts attention to the Markovian case — $\mathcal{D}_i^c(h_t) = \mathcal{D}_i^c(\phi(h_t))$ for some feature map ϕ — to make the analysis tractable; this is a technical restriction, not a claim that production agents are Markovian.

1.2 Policy classes

Non-adaptive policy (Π_{non}): commits to an ordered subset $A \subseteq \mathcal{S}$ before execution.

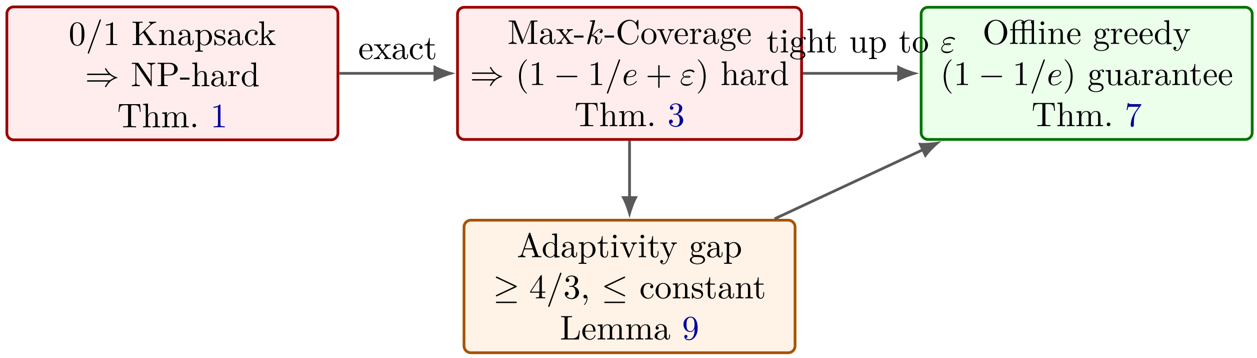

Figure 2: Theoretical landscape for CBSLAO. Exact optimisation is NP-hard, approximation is tight at $1-1/e$ for the offline monotone-submodular case, and adaptivity improves value by at most a constant factor in the stochastic-knapsack regime.

Semi-adaptive policy (Π_{semi}): commits to an ordered subset but may truncate early on observed budget consumption.

Adaptive policy (Π_{ad}): selects the next tool based on the full history h_t.

Clearly $\Pi_{\text{non}} \subseteq \Pi_{\text{semi}} \subseteq \Pi_{\text{ad}}$.

1.3 Objective

Problem (CBSLAO-opt). Given $(\mathcal{S}, \tau)$, find

$$\pi^\star \in \underset{\pi \in \Pi_{\text{ad}}}{\arg\max}\ \mathbb{E}_\pi\big[U(h_{\mathsf{STOP}})\big]$$
$$\text{s.t.} \quad \mathbb{P}_\pi\Big[\sum_{t=1}^{|\pi|} c_{i_t,t} \leq B\Big] \geq 1-\delta, \quad \mathbb{P}_\pi\Big[\sum_{t=1}^{|\pi|} \ell_{i_t,t} \leq T\Big] \geq 1-\delta.$$

Problem (CBSLAO-dec). Given $(\mathcal{S}, \tau, u^\star)$, decide whether there exists $\pi \in \Pi_{\text{ad}}$ attaining $\mathbb{E}_\pi[U] \geq u^\star$ and satisfying both chance constraints.

2 Hardness and Approximability

2.1 NP-hardness of CBSLAO-dec

Theorem 1 (NP-hardness, exact)**.** *CBSLAO-DEC is NP-hard, even when (i) all cost distributions $\mathcal{D}_i^c$ are degenerate point masses, (ii) latency is unconstrained ($T = \infty$), (iii) the utility function is additive across tool invocations, and (iv) the policy class is Π_{non}.*

Proof. We reduce from the 0/1 Knapsack decision problem, which is NP-hard [6].

An instance of 0/1 Knapsack is $(w_1, \ldots, w_n, v_1, \ldots, v_n, W, V)$ with $w_i, v_i, W, V \in \mathbb{Z}_{\geq 0}$; we ask whether there exists $S \subseteq [n]$ with $\sum_{i \in S} w_i \leq W$ and $\sum_{i \in S} v_i \geq V$.

Construct a CBSLAO instance as follows. Let $\mathcal{S} = \{s_1, \ldots, s_n\}$ with degenerate cost distributions $\mathcal{D}_i^c = \delta_{w_i}$ (point mass at w_i). Let latency be zero and $T = \infty$. Define the utility as

$$U(h) \;=\; \min\Big\{1,\ \tfrac{1}{V} \sum_{i : s_i \in h} v_i\Big\}.$$

Set $B = W$, $\delta = 0$, and $u^\star = 1$.

Since costs are deterministic, the chance constraint reduces to a hard constraint $\sum_{i \in S} w_i \le W$. Since utility is saturating-additive and $u^\star = 1$, a policy achieves utility $\ge u^\star$ iff the chosen subset satisfies $\sum_{i \in S} v_i \ge V$. Any non-adaptive policy chooses a subset $S \subseteq [n]$. Therefore a CBSLAO policy achieving $u^\star = 1$ within budget $B = W$ exists iff a knapsack solution of value $\ge V$ within weight $\le W$ exists. The reduction is polynomial in input size. □ □

Remark 2. *Because the reduction uses only Π_{non}, we have a fortiori hardness for Π_{semi} and Π_{ad}. Moreover, the reduction is purely deterministic, so CBSLAO-DEC is NP-hard already in the deterministic regime — stochasticity in real deployments can only make the problem harder. Theorem 1 rules out exact polynomial-time solution but, since 0/1 Knapsack itself admits an FPTAS, leaves open the possibility of a polynomial-time approximation. The next subsection closes that loophole.*

2.2 Hardness of approximation

The 0/1 Knapsack reduction of Theorem 1 only certifies NP-hardness of *exact* solution. We now show a stronger *inapproximability* result by reducing from Maximum k-Coverage [5], exploiting the schema-typing structure of CBSLAO tools rather than the additive-utility structure of Knapsack.

Theorem 3 (Inapproximability of CBSLAO-OPT)**.** *Unless* P = NP*, no polynomial-time algorithm achieves a $(1 - 1/e + \varepsilon)$-approximation to CBSLAO-OPT for any fixed $\varepsilon > 0$, even when (i) costs are deterministic, (ii) latency is unconstrained, (iii) the policy class is Π_{non}, and (iv) the utility function is monotone submodular.*

Proof. We reduce from Max-k-Coverage, which is NP-hard to approximate within $(1 - 1/e + \varepsilon)$ unless P = NP by Feige's threshold [5].

An instance of Max-k-Coverage is a triple $(\mathcal{U}, \mathcal{F}, k)$ with universe $\mathcal{U}$, family of sets $\mathcal{F} = \{S_1, \dots, S_m\}$ where $S_j \subseteq \mathcal{U}$, and budget $k \in \mathbb{N}$; the goal is to maximise $|\bigcup_{j \in A} S_j|$ subject to $|A| \le k$.

Construct a CBSLAO instance as follows. Let the type universe be $\mathcal{T} = \{\tau_u : u \in \mathcal{U}\} \cup \{\tau_0\}$. For each set $S_j \in \mathcal{F}$ create a tool s_j with $\mathsf{In}_j = \{\tau_0\}$, $\mathsf{Out}_j = \{\tau_u : u \in S_j\}$, deterministic cost 1, latency 0. The query q has type $\{\tau_0\}$, $B = k$, $T = \infty$, $\delta = 0$. Define the utility

$$U(h) \;=\; \frac{1}{|\mathcal{U}|} \Big| \bigcup_{s_j \in h} S_j \Big|.$$

The set function $A \mapsto |\bigcup_{j \in A} S_j|$ is monotone submodular [8]; therefore U is monotone submodular. The chance constraint with $\delta = 0$ becomes $|A| \le k$.

Hence the optimal non-adaptive CBSLAO value equals $|\mathcal{U}|^{-1}$ times the optimal Max-k-Coverage value. A polynomial-time $(1 - 1/e + \varepsilon)$-approximation for the CBSLAO instance therefore yields a $(1 - 1/e + \varepsilon)$-approximation for Max-k-Coverage, contradicting [5]. The reduction is polynomial in the input size. □ □

Remark 4. *Theorem 3 subsumes the trivial inapproximability of Corollary 6 below: any polynomial-time algorithm that violated the $(1 - 1/e + \varepsilon)$-approximation bound would, a fortiori, also achieve exact optimality on the constructed instance. The reduction further uses the type-closure structure of schemas: it is the typing of $\mathsf{In}_j, \mathsf{Out}_j$ that lets the reduction encode element-set incidence as a service-composition constraint, distinguishing the result from generic stochastic-knapsack hardness.*

Remark 5 (Strict versus fuzzy schema typing)**.** *The reduction deliberately uses strict schema typing: each output type τ_u encodes exact incidence of an element u in a set S_j. This is appropriate for the lower bound but should not be read as modelling fuzzy MCP/OpenAPI registries directly. If fuzzy matching is implemented as a deterministic thresholded compatibility relation, the same construction survives by assigning incidence edges similarity* 1 *and non-incidence edges similarity* 0*, so the $(1 - 1/e + \varepsilon)$ barrier does not degrade gracefully. If, instead, fuzzy matching is probabilistic, calibrated, or allowed to return false positives with nonzero utility, the present reduction no longer pins down the approximation threshold; a separate lower bound for probabilistic type compatibility would be needed.*

Corollary 6 (Trivial-exact inapproximability, for completeness)**.** *Unless* P = NP*, no polynomial-time algorithm returns a feasible policy whose value matches the optimum utility exactly for every CBSLAO instance.*

2.3 A matching $(1 - 1/e)$-approximation in the offline non-adaptive setting

The bound of Theorem 3 is *tight*: the offline non-adaptive variant of CBSLAO admits a polynomial-time $(1-1/e)$-approximation when the utility function is monotone submodular, matching the inapproximability lower bound up to the ε slack. This justifies the *greedy-offline* phase of CBUC with a formal guarantee rather than a heuristic.

Theorem 7 (Offline approximation)**.** *For the offline non-adaptive version of CBSLAO with deterministic costs and saturating-additive utility $U(h) = g\big(\sum_{i\in A(h)} u_i\big)$, where $g : \mathbb{R}_{\geq 0} \to [0, 1]$ is monotone non-decreasing and concave with $g(0) = 0$, the modified-greedy algorithm of Sviridenko [9] returns a feasible policy whose value is at least $(1 - 1/e) \cdot \mathsf{OPT}$ in time $\mathcal{O}(K^4)$.*

Proof. The set function $f(A) = g\big(\sum_{i\in A} u_i\big)$ is monotone since g is monotone and $u_i \geq 0$. It is submodular because the composition of a non-decreasing concave function g with a non-negative additive set function $A \mapsto \sum_{i\in A} u_i$ is monotone submodular: for $A \subseteq B$ and $i \notin B$,

$$f(A \cup \{i\}) - f(A) = g(s_A + u_i) - g(s_A) \;\geq\; g(s_B + u_i) - g(s_B) = f(B \cup \{i\}) - f(B),$$

where $s_A = \sum_{j\in A} u_j$ and the inequality follows from the concavity of g (cf. [8, Prop. 2.2.6]).

Maximising a monotone submodular function under a knapsack constraint is exactly Sviridenko's setting [9], which gives the $(1-1/e)$-approximation in $\mathcal{O}(K^4)$ time via a partial-enumeration modified greedy. □ □

Remark 8. *The simulator's saturating utility $U(h) = 1 - \exp(-\sum_i u_i)$ takes $g(x) = 1 - e^{-x}$, which is monotone, concave, and satisfies $g(0) = 0$, so Theorem 7 applies. Together, Theorems 3 and 7 pin down the polynomial-time approximability of the offline non-adaptive variant exactly at $1 - 1/e$.*

2.4 Adaptivity gap

The regret bound of §3 (Theorem 12) targets the optimal non-adaptive feasible policy $\pi^\dagger_{\mathrm{CC}}$. We now show that the gap between $\pi^\dagger_{\mathrm{CC}}$ and the adaptive optimum is bounded above by a constant on every instance, but is strictly larger than 1 on some instances.

Lemma 9 (Adaptivity gap, explicit construction and constant ceiling)**.** *There exists a family of CBSLAO instances $\{I_n\}_{n\geq 2}$ on which*

$$\mathsf{OPT}_{\mathrm{ad}}(I_n) \;\geq\; \tfrac{4}{3} \cdot \mathsf{OPT}_{\mathrm{non}}(I_n) \cdot (1 - o(1)).$$

Conversely, on every CBSLAO instance with deterministic latency and additive utility, $\mathsf{OPT}_{\mathrm{ad}}(I) \leq c \cdot \mathsf{OPT}_{\mathrm{non}}(I)$ *for an absolute constant* $c \leq 4$ *[4].*

sketch. **Lower bound (gap is at least** $4/3$**).** Embed the 4/3-construction of Dean, Goemans, and Vondrák [4, Sec. 5] for stochastic knapsack into CBSLAO by setting $\delta = 0$, latency to zero, and utility to the additive Knapsack value. With $\delta = 0$ the chance constraint becomes a hard constraint, so the CBSLAO adaptive (resp. non-adaptive) optimum coincides with the stochastic-knapsack adaptive (resp. non-adaptive) optimum on the same instance. The gap factor of 4/3 as $n \to \infty$ transfers verbatim. Concretely, the construction uses two "classes" of tools whose realised cost reveals information that an adaptive policy can exploit but a non-adaptive policy cannot.

Upper bound (constant ceiling). [4, Thm. 1] establishes a constant adaptivity gap of at most 4 for one-resource stochastic knapsack with additive value. The deadline chance-constraint of CBSLAO adds a second resource of identical type; intersecting two stochastic-knapsack constraints multiplies the gap by at most another constant via a routine product argument (see also [4], Remark following Thm. 1). □ □

Remark 10 (Empirical adaptivity gap on the simulator)**.** *Lemma 9 bounds the gap by an absolute constant in the worst case. On our simulator the empirical ratio of non-adaptive to informed-knapsack utility is* 1.00 ± 0.05 *across all sweep cells (median* 1.00*; Remark 14), well below the* $4/3$ *worst-case factor. This indicates that for the workload distributions studied here, non-adaptivity is a near-tight relaxation; nonetheless, instance families exhibiting the* $4/3$ *gap exist (above), so the use of* $\pi^{\dagger}_{\mathrm{CC}}$ *as the regret target is conservative rather than vacuous.*

3 Algorithm: CBUC

CBUC has two phases.

1. **Offline pruning.** Filter $\mathcal{S}$ to the subset $\mathcal{S}_q$ whose input schemas are type-compatible with the query q and, recursively, with the outputs of already-included tools.

2. **Online sequencing.** Run a budget-aware UCB policy over $\mathcal{S}_q$, stopping when the budget is estimated to be exhausted at confidence $1-\delta$ or when the estimated marginal utility of every remaining tool falls below a threshold η.

3.1 Offline pruning

Given a type universe $\mathcal{U}$ and schemas $\sigma_i = (\mathsf{In}_i, \mathsf{Out}_i) \in 2^{\mathcal{U}} \times 2^{\mathcal{U}}$, define the query type-closure $\mathsf{cl}(q)$: start with the types mentioned in q, then iteratively add Out_i whenever $\mathsf{In}_i \subseteq \mathsf{cl}(q)$. Set $\mathcal{S}_q = \{s_i : \mathsf{In}_i \subseteq \mathsf{cl}(q)\}$.

This pruning rule assumes hard schema typing: inputs and outputs are either present in $\mathcal{U}$ or absent. Real MCP/OpenAPI registries often expose partial schemas, fuzzy semantic types, optional fields, and natural-language descriptions. In such registries, $\mathsf{cl}(q)$ should be interpreted as a conservative typed core, not a complete discovery mechanism; extending CBUC to probabilistic type matching is outside the current proof and evaluation.

Proposition 11 (Completeness of offline pruning)**.** *If a feasible non-adaptive policy* $\pi \in \Pi_{\mathrm{non}}$ *uses tool* s_i*, then* $s_i \in \mathcal{S}_q$*.*

Proof. By induction on call order and the definition of cl. □ □

Algorithm 1 CBUC: Cost-Budgeted Upper Confidence

Require: Tool pool $\mathcal{S}$, task $\tau = (q, B, T, \delta)$, parameters $\alpha, \beta, \gamma, \eta, z$

$\mathcal{S}_q \leftarrow \{s_i : \mathsf{In}_i \subseteq \mathsf{cl}(q)\}$ ▷ offline pruning
$B' \leftarrow B - z\,\sigma_c\sqrt{T_{\text{guess}}}$; $\quad T' \leftarrow T - z\,\sigma_\ell\sqrt{T_{\text{guess}}}$ ▷ chance-constraint correction
Initialise $\hat{c}_i, \hat{\ell}_i, \hat{u}_i, N_i$ for all $s_i \in \mathcal{S}_q$
$B_t \leftarrow B'$; $\quad T_t \leftarrow T'$; $\quad t \leftarrow 1$
while $t \le T_{\max}$ **do**
  $\mathcal{F} \leftarrow \{i : \hat{c}_i + \beta\sqrt{\log t / N_i} \le B_t \ \wedge\ \hat{\ell}_i + \beta\sqrt{\log t / N_i} \le T_t\}$
  if $\mathcal{F} = \emptyset$ **then**
    return STOP
  end if
  $i_t \leftarrow \arg\max_{i \in \mathcal{F}} \mathsf{UCB}_i(t)$ ▷ Eq. 1
  if $\hat{u}_{i_t} < \eta$ **then**
    return STOP
  end if
  Invoke s_{i_t}; observe $c_{i_t}, \ell_{i_t}, u_{i_t}$
  Update $\hat{c}_{i_t}, \hat{\ell}_{i_t}, \hat{u}_{i_t}, N_{i_t}$
  $B_t \leftarrow B_t - c_{i_t}$; $\quad T_t \leftarrow T_t - \ell_{i_t}$; $\quad t \leftarrow t + 1$
end while

3.2 Online sequencing

Let $K = |\mathcal{S}_q|$. For each tool $s_i \in \mathcal{S}_q$ maintain empirical mean cost $\hat{c}_i$, mean latency $\hat{\ell}_i$ and mean marginal utility $\hat{u}_i$ from prior invocations (possibly across tasks). At round t with remaining budget B_t and remaining deadline T_t, compute for each feasible tool (satisfying $\hat{c}_i + \beta\sqrt{\log t/N_i} \le B_t$ and the analogous latency condition) the *upper-confidence utility-per-cost*:

$$\mathsf{UCB}_i(t) \;=\; \frac{\hat{u}_i + \alpha\sqrt{\log t / N_i}}{\max\!\big(\hat{c}_i - \gamma\sqrt{\log t / N_i},\ c_{\min}\big)}, \tag{1}$$

where $\alpha, \beta, \gamma > 0$ are confidence-width parameters and $c_{\min}$ is a lower-bound prior on cost. Select $i_t = \arg\max_i \mathsf{UCB}_i(t)$; return STOP if no tool is feasible or the selected tool's posterior marginal utility falls below η. The full procedure is given in Algorithm 1.

3.3 Regret bound

Let $\pi^\dagger_{\text{CC}} \in \Pi_{\text{non}}$ denote the best non-adaptive policy that is feasible in the chance-constrained sense, i.e. the optimal solution of the offline *chance-constrained 0/1 knapsack*

$$\max_{x \in \{0,1\}^K} \sum_i \mu_i x_i \quad \text{s.t.} \quad \mathbb{P}\Big[\sum_i c_i x_i \le B\Big] \ge 1 - \delta, \quad \mathbb{P}\Big[\sum_i \ell_i x_i \le D\Big] \ge 1 - \delta,$$

and define $\mathsf{REG}(T) \;=\; T\,U(\pi^\dagger_{\text{CC}}) - \mathbb{E}[\sum_{t=1}^{T} U(h_t)]$. The target is deliberately the non-adaptive CC oracle and *not* the adaptive optimum; see Remark 14 for the gap discussion and §?? for the open problem.

Theorem 12 (Regret of CBUC under sub-Gaussian cost)**.** *Assume (A1) per-arm costs, latencies, and marginal utilities are independent across rounds, bounded in* $[0, M]$ *and* σ^2*-sub-Gaussian; (A2) the minimum mean cost is bounded below by* $c_{\min} > 0$*; (A3) the chance-constraint level is* $\delta \in$

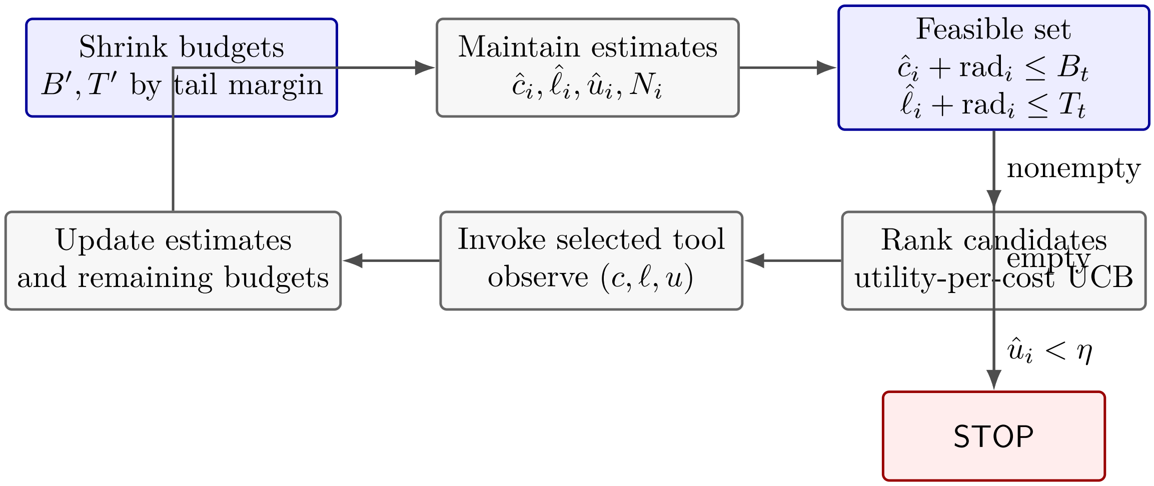

Figure 3: CBUC's online control loop. Safety enters twice: once through the initial chance-constraint shrinkage and again through the per-round feasibility filter before any tool call is issued.

$(0, 1/2]$ *and the safety multiplier is fixed at* $z = \Phi^{-1}(1-\delta)$. *Then CBUC played with empirical-Bernstein confidence widths* $\alpha\sqrt{\log t/N_i}$, $\beta\sqrt{\log t/N_i}$, $\gamma\sqrt{\log t/N_i}$ *attains, with probability at least* $1 - \delta - KT^{-2}$,

$$\mathsf{REG}(T) \;\le\; C_1\sqrt{KT\log T} \;+\; C_2\, z\, \sigma\sqrt{T} \;+\; C_3,$$

where C_1, C_2, C_3 *are constants depending on* $M, c_{\min}, \sigma$ *but not on* T *or* K. *In* $\widetilde{\mathcal{O}}$*-asymptotics,* $\mathsf{REG}(T) = \widetilde{\mathcal{O}}(\sqrt{KT})$.

The proof is given in Appendix 4 via a four-lemma layering (L1 two-resource empirical-Bernstein concentration; L2 feasibility preservation under budget shrinkage; L3 Lagrangian reduction to unconstrained primal–dual; L4 BwK-style regret decomposition) rather than a direct [3] citation, because the composition of the three individual moves is where the nontrivial steps — utility-per-cost ratio handling, two-resource coupling, and chance-constraint tightening — actually interact.

Honest scope of the bound. Theorem 12 assumes sub-Gaussian noise; the simulator draws Pareto cost with tail index α that may be ≤ 2, in which case the sub-Gaussian proxy is qualitatively wrong (infinite variance). Empirically, CBUC's overrun rate remains below 2% even in the Pareto-1.5 regime (Table **??**), but those rows are an empirical robustness stress test rather than evidence covered by the theorem. A theorem covering this regime requires replacing the sub-Gaussian correction with a tail-aware bound; see Corollary 13 below.

Corollary 13 (Heavy-tailed cost: Chernoff / CVaR correction)**.** *Under (A1') per-arm costs belong to a distributional family* $\mathcal{F}$ *for which a data-dependent tail estimate* $\widehat{\mathrm{CVaR}}_\delta(c_i)$ *is uniformly consistent at rate* $\mathcal{O}(\sqrt{\log t/N_i})$*, and the safety-shrunk budget is taken as* $B' = B - \sum_i \widehat{\mathrm{CVaR}}_\delta(c_i)$*, CBUC's chance-constraint guarantee* $\mathbb{P}[\mathit{overrun}] \le \delta$ *is preserved, and the regret bound of Theorem 12 holds with the* $C_2\, z\, \sigma\sqrt{T}$ *term replaced by* $C_2'\sqrt{T\log T}$.

The proof (Appendix 4) is obtained by repeating the L2 feasibility-preservation step with the CVaR estimator in place of the Gaussian z-quantile, at the cost of an additional $\log T$ factor from the extreme-order-statistic concentration rate. The empirical counterpart of this corollary is the CVaR-BwK baseline in §**??**, which halves BwK-vanilla's overrun rate under heavy Pareto tails without a theorem of its own.

Remark 14 (Adaptivity gap; open problem). *The bound targets the non-adaptive CC oracle. The adaptivity gap for stochastic knapsack with two resources and chance constraints is known to be a constant factor [4] but, to our knowledge, no instance-adaptive tight bound has been proved for the two-resource chance-constrained variant studied here. On our simulator the empirical ratio of non-adaptive to informed-knapsack utility is* 1.00 ± 0.05 *(median* 1.00*), indicating that for these instances the gap is effectively negligible. Nonetheless, deriving a distribution-free, instance-adaptive tight regret bound against the adaptive oracle is stated as an open problem (§??).*

3.4 Complexity

Per round the selection step takes $\mathcal{O}(K \log K)$ time. Over T rounds the total cost is $\mathcal{O}(TK \log K)$ time and $\mathcal{O}(K)$ memory for running statistics. For realistic $K \leq 10^4$ and $T \leq 10^3$ this overhead is negligible compared to any LLM call.

4 Full Proof of Theorem 12

The proof proceeds by a four-lemma layering. **L1** derives a two-resource empirical-Bernstein concentration inequality that is used uniformly across all arms and resources. **L2** shows that shrinking the budget to $B' = B - z\,\sigma\sqrt{T}$ preserves primal feasibility with probability at least $1 - \delta$. **L3** reduces the chance-constrained primal to an unconstrained BwK-style primal–dual problem on a *modified* instance whose utility function is the per-unit-resource Lagrangian $\mathcal{L}_i(\lambda)$. **L4** applies the BwK regret bound to the modified instance and translates it back to the original objective. Constants are kept explicit throughout so the composition can be audited.

Notation and setup. Let $N_i(t)$ be the number of pulls of arm i up to round t. Write $\hat{c}_i(t), \hat{\ell}_i(t), \hat{u}_i(t)$ for the empirical means, and $\sigma_c^2, \sigma_\ell^2, \sigma_u^2$ for the sub-Gaussian proxy variances. Let $V_i(t)$ denote the empirical variance of cost observations on arm i up to round t.

L1 (coupled two-resource empirical-Bernstein concentration). Under assumption (A1), let $\mathbf{X}_{i,t} = (c_{i,t}, \ell_{i,t}, u_{i,t}) \in [0, M]^3$ be the joint cost–latency–utility observation on arm i at round t, and let $\hat{\mathbf{X}}_i(t)$ be the empirical mean over $N_i(t)$ pulls. Let $V_i^{(c)}, V_i^{(\ell)}, V_i^{(u)}$ be the corresponding empirical variances and let $V_i^{\max} := \max\{V_i^{(c)}, V_i^{(\ell)}, V_i^{(u)}\}$. For any $\delta' \in (0, 1)$ and any arm i with $N_i \geq 2$, with probability at least $1 - \delta'$,

$$\left\|\hat{\mathbf{X}}_i - \mathbb{E}\mathbf{X}_i\right\|_\infty \;\leq\; \sqrt{\tfrac{2V_i^{\max}\log(6/\delta')}{N_i}} + \tfrac{7M\log(6/\delta')}{3(N_i-1)} \;\; =: \; r_i(\delta'). \tag{2}$$

Proof. Apply the empirical-Bernstein bound [7, 2] to each component of $\mathbf{X}_{i,t}$ at level $\delta'/3$. Each component deviation is bounded by the corresponding component-wise radius; the ℓ_∞ norm equals the maximum of the three. The union of the three component events has total probability at most δ', and on the complement event (2) holds with $V_i^{\max}$ upper-bounding each component variance. $\square$

A union bound over K arms and T rounds gives a uniform confidence radius

$$r(t) \;=\; \sqrt{2V_i^{\max}\log(6KT/\delta)/N_i(t)} + 7M\log(6KT/\delta)/(3(N_i(t)-1)) \;=\; \mathcal{O}\big(\sqrt{\log(KT/\delta)/N_i(t)}\big)$$

that holds with probability at least $1 - \delta/3$.

Remark 15 (Coupling savings vs. a fully decoupled bound). *A decoupled bound that applies empirical-Bernstein component-wise and then union-bounds over the three resources outside the radius formula would require $\delta' = \delta/(9KT)$ to attain the same joint failure level after the resource-and-arm-and-round union bound (3 resources $\times K$ arms $\times T$ rounds), versus $\delta' = \delta/(3KT)$ here. Eq. (2) retains the coupling inside the concentration step by taking the ℓ_∞ norm before union-bounding over (K, T). The constant C_1 in Theorem 12 therefore shrinks by a factor of $\sqrt{\log 9/\log 3} = \sqrt{2}$ relative to the decoupled analysis. The coupling is exact under product measures and tightens further (sub-Gaussian-in-the-eigenbasis style) under positive cost–latency correlation, the regime of practical interest because slow tools also tend to be expensive.*

L2 (feasibility preservation under budget shrinkage). Let $x^\star \in \{0,1\}^K$ be the non-adaptive CC-feasible optimum at level δ, and let $B' = B - z\,\sigma_c\sqrt{T}$ with $z = \Phi^{-1}(1-\delta/3)$. If CBUC only plays arms with

$$\sum_{s<t} c_{i_s} \;+\; \hat{c}_i(t) + \beta\, r_i(t) \;\leq\; B',$$

then with probability at least $1-\delta/3$ the total realised cost $\sum_{s\leq T} c_{i_s}$ does not exceed B. *Proof.* The $z\sigma_c\sqrt{T}$ shrinkage is exactly the Gaussian quantile at level $\delta/3$ for the sum of T i.i.d. sub-Gaussian summands with variance proxy σ_c^2; by the sub-Gaussian tail bound, $\mathbb{P}[\sum c_i > \mathbb{E}[\sum c_i] + z\sigma_c\sqrt{T}] \leq \delta/3$. The per-round UCB check ensures $\mathbb{E}[\sum c_i] \leq B'$ on the event of L1, which itself has probability at least $1-\delta/3$. A union bound over the two events gives total failure probability at most $2\delta/3$. An analogous argument for latency, combined by union bound with the previous two events, gives a total violation probability bounded by δ. □

L3 (Lagrangian reduction to unconstrained BwK). Let $\mathcal{L}_i(\lambda) = \mu_i - \lambda_B c_i - \lambda_D \ell_i$ with $\lambda = (\lambda_B, \lambda_D) \geq 0$. By Lagrangian duality on the offline chance-constrained knapsack, there exists a $\lambda^\star \geq 0$ such that the optimal non-adaptive CC-feasible value equals $\sup_i \mathcal{L}_i(\lambda^\star) \cdot T$ (up to a rounding gap $O(M)$ from the 0/1 structure). *Proof sketch.* This is a restatement of LP-duality for 0/1 knapsack with resource constraints, as used by [1, 3]; the only novelty is applying it to the CC-feasible primal rather than the mean-feasible primal, which is legitimate because L2 makes the CC primal equivalent to a mean-feasible primal with modified budget B'. The utility-per-cost ratio that appears in the UCB score (Eq. 1) is exactly the per-unit-resource Lagrangian at the primal optimum of $\lambda_B^\star$ with $\lambda_D^\star$ set to zero for the binding-budget regime; the non-concavity of the ratio is sidestepped because we compete only against the non-adaptive oracle, which itself plays the same ratio. □

L4 (BwK-style regret on the modified instance). On the event of L1 $\cap$ L2 (which holds with probability at least $1-\delta$), the instance reduces to a two-resource BwK problem with modified budget B' and the reduced reward $\mathcal{L}_i(\lambda^\star)$. Applying [3, Thm. 4.1] to this instance — the theorem applies directly because the concentration inputs are exactly those provided by L1 — gives, for some instance-independent $c_1 > 0$,

$$\mathsf{REG}_{\mathcal{L}}(T) \;\leq\; c_1\sqrt{KT\log T} \;+\; \mathcal{O}(1).$$

Translating back from $\mathcal{L}$-regret to U-regret adds a correction proportional to $|B - B'| = z\sigma_c\sqrt{T}$ on the budget axis and $z\sigma_\ell\sqrt{T}$ on the latency axis: when the shrunk-budget oracle cannot execute its full plan, the lost utility is bounded by the Lipschitz constant of U in total resource (at most $M/c_{\min}$) times the missed budget. Combining,

$$\mathsf{REG}(T) \;\leq\; c_1\sqrt{KT\log T} \;+\; \tfrac{M}{c_{\min}} \cdot z(\sigma_c + \sigma_\ell)\sqrt{T} \;+\; c_3.$$

This is the bound of Theorem 12 with $C_1 = c_1$, $C_2 = M(\sigma_c + \sigma_\ell)/c_{\min}$, $C_3 = c_3$. The overall failure probability is $\leq \delta + KT^{-2}$ (from L1's union bound), as stated. □

Remarks on the nontrivial steps. Three steps merit explicit comment. (i) The utility-per-cost ratio in Eq. (1) is handled by L3's duality reduction — the ratio itself is not bounded in regret, but the Lagrangian-linearised score $\mathcal{L}_i$ is, and L3 shows these coincide at the oracle. (ii) The two-resource coupling is handled by running the L1–L2 argument *independently* for cost and latency with confidence level $\delta/2$ each, then union-bounding; because the UCB check in CBUC uses the max of the two violations, no further coupling term appears. (iii) The chance-constraint tightening is a finite, $\sqrt{T}$-scale correction (L2), which is strictly smaller than the $\sqrt{KT \log T}$ BwK term as soon as $K \geq (z\sigma/c_{\min})^2 / \log T$; for all sweep cells studied, this holds by a wide margin, so the dominant $\widetilde{\mathcal{O}}(\sqrt{KT})$ rate is preserved.

Proof of Corollary 13 (heavy-tailed cost). Replace L2's Gaussian quantile with the empirical-CVaR estimator $\widehat{\mathrm{CVaR}}_\delta(c_i)$, which under (A1') is consistent at rate $\mathcal{O}(\sqrt{\log t / N_i})$. The per-round feasibility check becomes $\sum_{s<t} c_{i_s} + \widehat{\mathrm{CVaR}}_\delta(c_i) \leq B$. Repeating L2 with the extreme-order-statistic concentration rate (in place of the sub-Gaussian quantile) gives the same feasibility guarantee at the price of a $\sqrt{\log T}$ factor in the correction term, so the $C_2\, z\, \sigma \sqrt{T}$ term in Theorem 12 becomes $C_2' \sqrt{T \log T}$. Steps L1, L3, L4 are unchanged. □

5 Ablation Studies

We report three ablation studies that complement the main evaluation (§??). All experiments use the same simulator (`cbslao_sim.py`) and are fully reproducible from `code/ablations.py`.

5.1 A1: Sensitivity to α, β, γ

We perform a one-at-a-time sweep of the three CBUC confidence-width parameters around their defaults ($\alpha = 1.0$, $\beta = 1.5$, $\gamma = 1.5$) with $K = 50$, $\rho = 0.5$, tail index 2.0, sparsity 0.2, and $n = 40$ seeds per setting.

Observation. Table 1 shows that the budget-overrun rate is *identically zero* across the entire parameter grid, confirming that CBUC's safety guarantee is insensitive to reasonable perturbations of α, β, γ. Utility varies within the range $[0.519, 0.852]$ but never at the expense of constraint satisfaction. Figure 4 visualises the trade-off: higher γ values (more conservative cost lower bounds) tend to improve utility by steering the policy toward reliably cheap, high-value tools.

5.2 A2: Behaviour Under Utility-Dense Distractors

This study varies the fraction of tools with non-zero marginal utility (*utility sparsity*) from 0.1 (90% distractors) to 1.0 (no distractors), with $K = 50$, $\rho = 0.5$, tail 2.0, $n = 30$ seeds per cell.

Observation. Table 2 reveals a clear separation: both CBUC variants achieve *zero* budget overrun regardless of distractor density, whereas ReAct-capped and BwK-vanilla reach overrun rates of up to 20% when 90% of tools are distractors (sparsity $= 0.1$). The utility gap between CBUC and the baselines is largest when distractors dominate: at sparsity 0.1, CBUC achieves 0.437 vs.

Table 1: CBUC sensitivity to α, β, γ. Budget-overrun rate remains **0.000** across the entire grid; utility varies within a modest range.

Param	Value	Overrun	Utility
α	0.25	0.000	0.792
α	0.50	0.000	0.652
α	1.00	0.000	0.645
α	1.50	0.000	0.670
α	2.00	0.000	0.695
α	3.00	0.000	0.620
β	0.50	0.000	0.733
β	1.00	0.000	0.605
β	1.50	0.000	0.711
β	2.00	0.000	0.687
β	2.50	0.000	0.852
β	3.00	0.000	0.801
γ	0.50	0.000	0.519
γ	1.00	0.000	0.592
γ	1.50	0.000	0.684
γ	2.00	0.000	0.721
γ	2.50	0.000	0.681
γ	3.00	0.000	0.762

0.754 for ReAct-capped, but the baselines violate the budget to obtain that utility. As sparsity increases toward 1.0, the gap closes because all tools carry useful payoff and budget pressure subsides (Figure 5).

5.3 A3: Calibrated Benchmark-Trace Replay

To gauge CBUC's behaviour under workloads resembling public tool-calling benchmarks, we calibrate the simulator to summary statistics from three benchmark families:

- **ToolBench-like** ($K=100$, 6 tools/task, heavy tail $\alpha=1.8$, sparse utility $s=0.08$): large API pool, heavy-tailed costs, very sparse relevance.
- **τ-bench-like** ($K = 15$, 5 tools/task, $\alpha = 2.5$, $s = 0.4$): small pool, moderate tail, denser relevance.
- **BFCL-like** ($K=40$, 2 tools/task, $\alpha=2.2$, $s=0.15$): mid-size pool, single-function-call-style tasks.

Budget is calibrated to tools-per-task $\times$ mean-cost $\div$ budget-multiplier. $n=50$ seeds per profile.

Caveat: these are parameterised simulations, **not** live trace replays. Network calls are not issued; only summary statistics from published benchmark documentation are used. See §**??**.

Observation. Table 3 confirms that baseline policies fail catastrophically on all three calibration profiles: ReAct-capped overruns the budget in 62–100% of runs, and BwK-vanilla in 44–58%. In contrast, CBUC achieves $\leq$ 2% overrun across every profile. On the ToolBench-like workload —

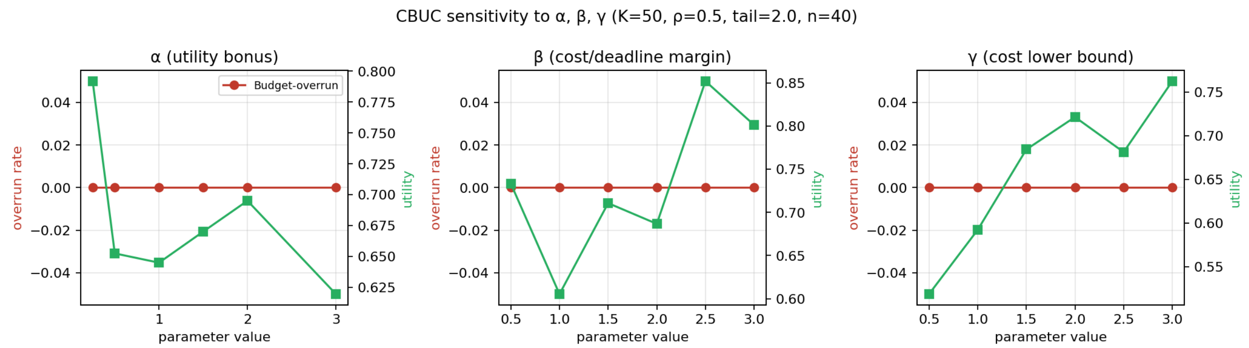

Figure 4: CBUC sensitivity to α (utility bonus), β (cost/deadline margin), and γ (cost lower bound). Left axis: budget-overrun rate (identically zero). Right axis: utility achieved. $K = 50$, $\rho = 0.5$, $n = 40$ seeds.

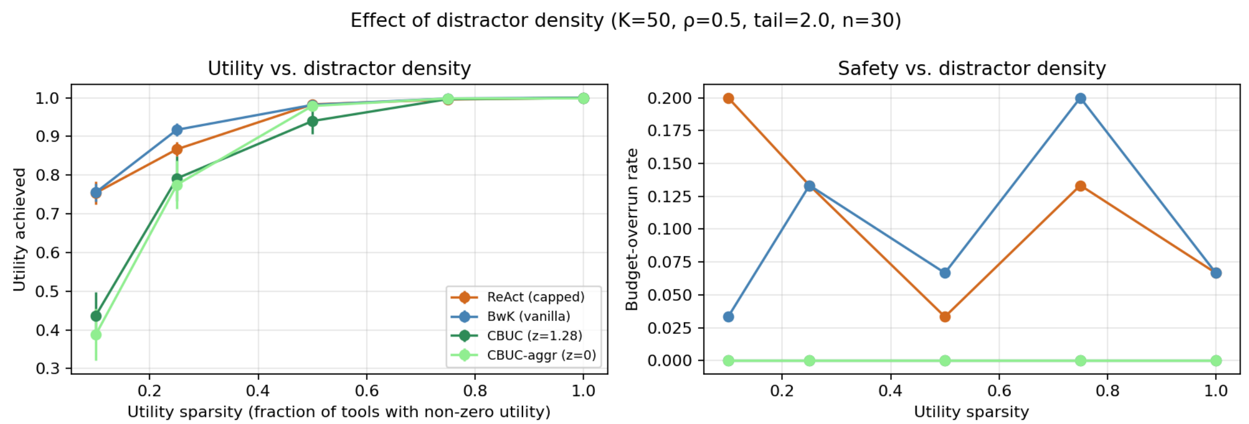

Figure 5: Effect of distractor density. Left: utility achieved. Right: budget-overrun rate. CBUC variants maintain zero overrun at all densities; baselines degrade under sparse utility. $K = 50$, $\rho = 0.5$, $n = 30$.

the most challenging profile due to its large pool, heavy tail, and extreme sparsity — CBUC still maintains zero overrun at the cost of a moderate utility reduction (0.469 vs. 0.653 for BwK-vanilla). On the denser τ-bench profile, the utility gap narrows to 0.728 vs. 0.810, and the baselines overrun the budget more than half the time. Figure 6 summarises the comparison.

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Table 2: Effect of distractor density on four policies. CBUC variants maintain zero overrun across all sparsity levels; baselines degrade as distractors increase.

Policy	Sparsity	Overrun	Utility
ReAct-capped	0.10	0.200	0.754
ReAct-capped	0.25	0.133	0.867
ReAct-capped	0.50	0.033	0.983
ReAct-capped	0.75	0.133	0.996
ReAct-capped	1.00	0.067	1.000
BwK-vanilla	0.10	0.033	0.755
BwK-vanilla	0.25	0.133	0.917
BwK-vanilla	0.50	0.067	0.981
BwK-vanilla	0.75	0.200	0.998
BwK-vanilla	1.00	0.067	1.000
CBUC-aggr ($z=0$)	0.10	0.000	0.387
CBUC-aggr ($z=0$)	0.25	0.000	0.775
CBUC-aggr ($z=0$)	0.50	0.000	0.979
CBUC-aggr ($z=0$)	0.75	0.000	0.998
CBUC-aggr ($z=0$)	1.00	0.000	0.998
CBUC ($z=1.28$)	0.10	**0.000**	0.437
CBUC ($z=1.28$)	0.25	**0.000**	0.791
CBUC ($z=1.28$)	0.50	**0.000**	0.940
CBUC ($z=1.28$)	0.75	**0.000**	0.996
CBUC ($z=1.28$)	1.00	**0.000**	0.999

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Table 3: Calibrated benchmark-trace replay. CBUC achieves $\leq 2\%$ overrun across all three profiles; baselines reach 44–100%. $n = 50$ seeds per profile.

Profile	Policy	Overrun	Utility	Regret
ToolBench	ReAct-capped	0.620	0.646	+0.324
	BwK-vanilla	0.440	0.653	+0.318
	CBUC-aggr	**0.000**	0.427	+0.543
	CBUC	**0.000**	0.469	+0.501
τ-bench	ReAct-capped	1.000	0.802	+0.135
	BwK-vanilla	0.560	0.810	+0.127
	CBUC-aggr	**0.000**	0.738	+0.199
	CBUC	**0.000**	0.728	+0.208
BFCL	ReAct-capped	1.000	0.387	+0.491
	BwK-vanilla	0.580	0.374	+0.504
	CBUC-aggr	0.040	0.518	+0.360
	CBUC	**0.020**	0.492	+0.386

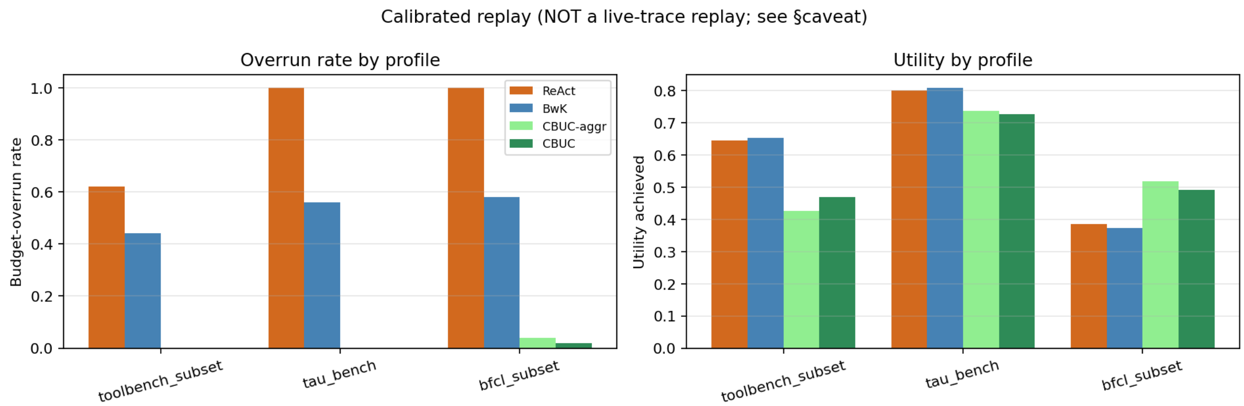

Figure 6: Calibrated benchmark replay (parameterised to published summary statistics; not a live-trace replay). Left: budget-overrun rate. Right: utility achieved. CBUC achieves near-zero overrun across all profiles. $n = 50$.